

Benchmarking Deep-Reinforcement-Learning Anti-Jamming Against Cryptographic Frequency Hopping for GPS-Denied Drone Positioning

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Abstract—Drones increasingly operate where the Global Positioning System (GPS) is jammed or spoofed, and Time-Difference-of-Arrival (TDOA) multilateration from ground beacons is a proven GPS-independent alternative whose reliability depends entirely on the radio channel carrying its positioning signals. We reproduce, train, and scale a deep-Q-network (DQN) anti-jamming model and benchmark it, under an identical environment, jammer set, and metric suite, against a training-free cryptographic frequency-hopping baseline and classical spread spectrum baselines, for the task of securing drone TDOA positioning. All methods, a ten-class jammer taxonomy spanning narrowband to wideband barrage, and released labelled spectrum-occupancy datasets run in a self-contained, CPU-only simulator with fixed seeds. The maximum-likelihood TDOA estimator attains the Cramér–Rao lower bound in line-of-sight (efficiency ≈ 1.0), and RANSAC mitigation cuts non-line-of-sight median error up to $20\times$. The learned model matches or beats cryptographic hopping at small channel counts (avoidance 0.91 vs 0.70 at $M=10$) but degrades as channels scale toward military-realistic counts, under both an equal-wall-clock and an equal-gradient-step training regime, whereas cryptographic hopping is unaffected and reaches avoidance ≈ 1.00 against an intelligent adversary that drives predictable hopping to 0.0. A valid message signature still loses to a meaconing replay (attack success 1.0); an authenticated-timing layer (HMAC + nonce + time-of-flight) drives spoofing and replay success to 0.0. Per decision, cryptographic hopping costs 13 pJ versus up to 9.3 μ J for deep-learning inference ($\sim 7.2 \times 10^5\times$), and needs no training. We release the datasets, code, and benchmark.

Index Terms—GPS-denied positioning, TDOA multilateration, cryptographic frequency hopping, deep reinforcement learning, anti-jamming, replay attacks, reproducible benchmark.

I. INTRODUCTION

UNMANNED aerial vehicles now perform safety-critical roles, inspection, delivery, search-and-rescue, and defence, that demand reliable positioning even when an adversary is actively denying it. GPS signals are weak and unauthenticated, so low-cost jammers can deny them [18] and spoofers can inject a false location [24], [25], [19]. A robust, GPS-independent capability is therefore essential. TDOA multilateration provides one: a drone measures the

differences in arrival time of signals from beacons at known positions and solves a hyperbolic system for its own location, requiring neither GPS nor an on-board clock [6], [8], [30]. Its resilience, however, is only as strong as the radio channel carrying the positioning signals. If those signals are jammed no fix is possible, if a beacon is spoofed the fix is wrong, and if a genuine signal is recorded and replayed with a delay the timing that TDOA relies upon is silently corrupted. Securing the channel is thus inseparable from securing the position.

Two communities offer competing answers. Academic work increasingly applies deep learning to detect jamming and adaptively hop to clean channels [11], [12], [13], [31]. Fielded systems that must survive real jamming instead use cryptographically-driven frequency hopping, in which a shared secret key generates an unpredictable hop sequence [1], [2], [17]. The learned approach is data-driven, adaptive, and avoidance-based; the cryptographic approach is proven, training-free, and unpredictability-based. No prior work has placed the two side by side, under an identical environment, for drone positioning, nor used cryptographic hopping to secure a TDOA position estimate against the full jam/spoof/replay threat.

A. Problem Statement

Let the drone position be $p \in \mathbb{R}^3$ and let beacons $s_0, \dots, s_{N-1} \in \mathbb{R}^3$ be known. With $r_i = \|p - s_i\|$, the measured range differences relative to reference beacon 0 are

$$d_i = (r_i - r_0) + n_i, \quad i = 1, \dots, N-1, \quad (1)$$

where n_i is estimation noise; the position estimate is the maximum-likelihood solution

$$\hat{p} = \arg \min_p (d - h(p))^T \Sigma^{-1} (d - h(p)), \quad (2)$$

with $h_i(p) = \|p - s_i\| - \|p - s_0\|$. The positioning signals are carried on one of M channels chosen each slot t by a hopping policy c_t ; a jammer denies a set $J_t \subseteq \{0, \dots, M-1\}$. The security objective is to maximise the jamming-avoidance rate

$$A = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[c_t \notin J_t] \quad (3)$$

while keeping the estimate (2) authentic under spoofing and replay, and while remaining tractable as M grows toward the 2,320-channel configuration of fielded frequency-hopping radios [17].

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The labelled spectrum-occupancy and TDOA datasets, the simulator, and all scripts needed to reproduce every figure and table in this paper are available at <https://github.com/Ahmad-prog/secure-tdoa-benchmark> under the MIT licence.

B. Contributions

- 1) An open, reproducible benchmark with released labelled spectrum-occupancy and TDOA datasets, a ten-class jammer taxonomy spanning narrowband through wideband barrage, and a single simulator hosting every method so that the hopping strategy is the only variable that changes.
- 2) The first controlled head-to-head of cryptographic hopping against a reproduced DQN anti-jamming model under one environment and metric suite.
- 3) An empirical model-scaling characterisation under two training regimes, equal wall clock and equal gradient steps, separating degradation caused by a compute budget from degradation intrinsic to the learning problem.
- 4) An authenticated-timing layer that closes the replay/meaconing gap left open by hopping and by message signatures alike, integrated with rather than replacing an existing protocol stack.
- 5) A per-decision cost and energy analysis, and an explicit account of the operational limits of the cryptographic approach.

The remainder of the paper is organised as follows. Section II reviews related work; Section III defines the system, hopping strategies, and threat model; Section IV gives the learned model, key management, synchronisation, and the authenticated-timing layer; Section V describes the experimental setup; Section VI reports results; Section VII discusses boundary conditions and limitations; Section VIII concludes.

II. RELATED WORK

A. TDOA positioning for GPS-denied UAVs

Hyperbolic multilateration from arrival-time differences is a mature localisation principle: the Chan–Ho closed form [6] gives a near-maximum-likelihood estimate, Foy’s Taylor-series method [8] refines it, and GCC-PHAT [7] estimates the delays; the CRLB bounds the achievable accuracy [27], [29], [30], and recursive filters track the trajectory [26]. Recent work applies TDOA to UAVs and characterises altitude, bandwidth, and NLOS bias [9], [10], [15]. *Limitation.* This line assumes a benign channel and does not secure the positioning waveform against an active jamming, spoofing, or replay adversary.

B. Deep-learning anti-jamming

Deep reinforcement learning and convolutional models learn to sense the spectrum and hop to clean channels [11], [12], [13], [31], building on DQN control [35] and learned RF signal processing [33], [34]. *Limitation.* A learned policy needs labelled data and training, is predictable to an intelligent adversary, and its action space grows with the channel count, so scalability to wide-band systems is unaddressed. We reproduce the DQN with coarse-grained spectrum prediction of Zhang, Wu and Hu [11] as the learned arm of the comparison, since it specifies its ten-channel environment and jammer types completely enough to re-implement.

C. Cryptographic and spread-spectrum hopping

Cryptographic frequency hopping and spread spectrum are decades-old, proven anti-jamming techniques [1], [4], [5], [32]; uncoordinated and keyed spread spectrum provide jamming resistance and key establishment [2], [16], and follower-jamming limits are known [3]. *Limitation.* These are studied as communication primitives; none is benchmarked against a learned model under one environment, nor applied to secure a TDOA *position* estimate. Cryptographic hopping is adopted here unchanged, as a proven reference point rather than a contribution.

D. GPS spoofing and meaconing

Spoofing and meaconing (delayed replay) are well-documented GNSS threats [24], [25], and machine learning has been used to detect spoofing on UAVs [14]; distance-bounding [23] and message authentication [21] are relevant primitives. *Limitation.* A valid content signature does not protect arrival-time integrity, so replay can corrupt TDOA timing without forging anything, a gap not closed by prior work.

III. SYSTEM AND THREAT MODEL

A. Range noise and the CRLB

The per-range estimation standard deviation follows the time-of-arrival bound [27]

$$\sigma_r = c\sigma_\tau, \quad \sigma_\tau = \frac{1}{2\sqrt{2}\pi\sqrt{\text{SNR}}\beta}, \quad \beta = \frac{B}{\sqrt{3}}, \quad (4)$$

with c the speed of light, B the bandwidth, and β the RMS bandwidth. The range-difference covariance is $\Sigma = \sigma_r^2(I + \mathbf{1}\mathbf{1}^\top)$, and with Jacobian $G_i = u_i - u_0$ (u_k the unit vector to beacon k) the position CRLB is

$$\text{CRLB}(p) = \sqrt{\text{tr}[(G^\top \Sigma^{-1} G)^{-1}]}, \quad \eta = \frac{\text{CRLB}}{\text{RMSE}}. \quad (5)$$

The estimator solves (2) by a Chan–Ho initialiser [6] followed by damped (Levenberg–Marquardt) Taylor refinement [8]; under NLOS a RANSAC consensus over beacon subsets [28], [38] rejects biased beacons.

B. Hopping strategies

Both ends derive the channel from an AES-128 keystream [20], [22] under the shared key κ and a synchronised counter,

$$c_t = (\text{AES}_\kappa(t) \bmod 2^{32}) \bmod M, \quad (6)$$

which is statistically uniform and unpredictable to any adversary lacking κ . Direct-sequence spread spectrum (DSSS) and uncoordinated spread spectrum [2] provide the spread-spectrum baselines, and a public pseudo-random hopper and a fixed channel model keyless, predictable strategies. DSSS is modelled at the link level by its processing-gain margin: it delivers full throughput while the jammed fraction of the band stays below a tolerance of 0.34, and degrades beyond it.

C. Jammer taxonomy and threat model

Ten jammer classes span three tiers. Tier 1 is non-adaptive (constant, random channel, pulsed, and wideband barrage); Tier 2 is patterned (sweep, comb, partial-band); Tier 3 is intelligent (reactive follower, predictive learning jammer, smart partial-band). The barrage class denies a fresh random 70% of the band each slot and is the only class that exceeds the DSSS processing-gain margin; every narrowband class sits below it. The adversary may additionally inject forged beacon packets (spoofing) or capture and re-transmit genuine packets with an added delay (meaconing replay), the latter corrupting arrival-time integrity without forging any content.

IV. METHOD

A. Learned anti-jamming model

The DQN observes a sliding window of the per-channel occupancy, $s_t \in \{0, 1\}^{W \times M}$, and estimates $Q(s_t, a; \theta)$ with one output per channel; it acts greedily $c_t = \arg \max_a Q$ and is trained on the immediate reward $r_t = 1[c_t \notin J_t]$ (one-step / coarse-grained spectrum prediction, $\gamma=0$) by minimising $(Q(s_t, c_t; \theta) - r_t)^2$. The output layer has M units, so the parameter count and per-decision cost grow as $O(M)$.

B. Key management

The hop sequence in (6) is only as available as its key. The baseline threat model assumes pre-shared, pre-provisioned keys loaded before the mission, which is standard practice for keyed frequency hopping and analogous to crypto-fill in fielded radios; no over-the-air exchange is then required. Where a key must be established without a pre-shared secret under active jamming, jamming-resistant key establishment via uncoordinated frequency hopping [16] applies, and authenticated Diffie–Hellman covers a secure pre-flight channel. Rekeying is driven by the same counter, yielding fresh subkeys per epoch. The security of the scheme rests on key secrecy, which we state as an explicit assumption and revisit in Section VII.

C. Hop synchronisation

Correct de-hopping requires beacon and drone to agree on the current hop index. Synchronisation is ensured at three levels. First, TDOA already requires the beacons to share a common, disciplined timebase, since they are fixed infrastructure at known coordinates, so the hop clock reuses that timebase and introduces no new synchronisation burden. Second, the drone acquires initial synchronisation through an acquisition/sync-word (late-entry) sequence and thereafter advances its counter locally. Third, periodic sync-refresh slots and a tolerance window absorb clock drift and time-of-flight offset. The tolerance is quantified in Section VI as a function of hop rate.

Algorithm 1 Authenticated-timing verification at the drone

Require: packet (id, T_{tx} , nonce, tag), key κ , arrival T_{rx} , geometric delay δ , issued nonce ν , tolerance $\tau = 3\sigma_r/c$

Ensure: accept / reject

- 1: **if** tag $\neq$ HMAC $_{\kappa}(\text{id} \parallel T_{tx} \parallel \text{nonce})$ **then**
- 2: **return** reject $\triangleright$ forged or altered
- 3: **end if**
- 4: **if** nonce $\neq \nu$ **then**
- 5: **return** reject $\triangleright$ stale / replayed
- 6: **end if**
- 7: **if** $|T_{rx} - (T_{tx} + \delta)| > \tau$ **then**
- 8: **return** reject $\triangleright$ delayed re-transmission
- 9: **end if**
- 10: **return** accept

D. Authenticated timing

Cryptographic hopping defeats jamming (unpredictable sequence) and spoofing-by-injection (wrong channel and no valid tag) but not replay: an attacker re-transmits a genuine packet with an added delay, corrupting the TDOA timing without forging anything. The layer in Algorithm 1 composes three standard primitives, an HMAC [21], a freshness nonce, and a time-of-flight plausibility test in the manner of distance bounding [23], for the specific purpose of protecting a *time-of-arrival* measurement rather than message content. It is deployed alongside whatever authentication an existing protocol stack already provides and replaces none of it; Section VI quantifies why the stack's own signature is not sufficient on its own.

E. Evaluation metrics

The jamming-avoidance rate A is given by (3); positioning error is $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i \|\hat{p}_i - p_i\|^2}$ with efficiency η from (5); attack success is $\rho = \frac{1}{K} \sum_k \mathbf{1}[\text{attack}_k \text{ accepted} \wedge \text{harmful}]$; normalized throughput is the per-slot delivered fraction; and per-decision energy is $E = \text{MACs} \times e_{\text{MAC}}$ [39] for the DQN versus one AES block for cryptographic hopping.

V. EXPERIMENTAL SETUP

The simulator is self-contained and CPU-only (Python, NumPy/SciPy, the `cryptography` library); the DQN is a from-scratch NumPy network (window $W=16$, hidden 64, Adam, replay 2000, batch 32), which makes its $O(M)$ scaling cost a measured quantity rather than an assertion. Positioning uses $N=8$ ground beacons on a 1 km hexagon with mast-height diversity and a drone at ~ 300 m. Spectrum episodes are 500 slots; anti-jam results average 30 seeds, the equal-wall-clock scaling arm 10 seeds, and the equal-gradient-step arm 3 seeds; every run is reproducible from a fixed seed. Channel scales are $\{10, 100, 1000, 2320\}$, the last being the SINCGARS 30–88 MHz/25 kHz configuration [17]. The labelled spectrum-occupancy dataset (Fig. 1) is released with the code.

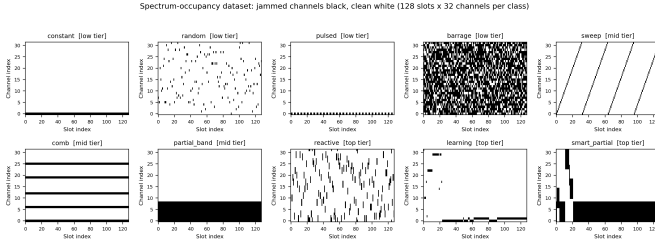


Fig. 1. Released spectrum-occupancy dataset: per-slot channel state (jammed in black) for the ten jammer classes; over a sliding window this is the DQN input.

TABLE I
TDOA POSITIONING (METRES). LOS REACHES THE CRLB; RANSAC MITIGATION CUTS NLOS MEDIAN ERROR UP TO 20 $\times$.

SNR (dB)	LOS RMSE	CRLB	NLOS median	NLOS + mitig.
10	26.8	26.6	138.4	67.8
20	7.8	8.2	133.2	33.3
30	2.7	2.6	147.7	7.4

VI. RESULTS

A. Positioning accuracy against the CRLB

The maximum-likelihood estimator attains the CRLB in LOS across the SNR range (efficiency $\eta \approx 0.97$ –1.0; RMSE 7.8 m vs CRLB 8.2 m at 20 dB), and a signal-level GCC-PHAT check validates the noise model (empirical/analytic σ_r ratio 1.03). NLOS error is heavy-tailed, so we report the median: RANSAC mitigation reduces it from 133 m to 33 m at 20 dB and from 148 m to 7.4 m at 30 dB (Table I, Fig. 2).

B. Anti-jamming across the taxonomy

Against the ten-class taxonomy, cryptographic hopping avoids collisions with a rate that rises with M (mean avoidance 0.74 $\rightarrow$ 0.87) and, crucially, resists the intelligent adversary: against the learning jammer its avoidance is 0.91 $\rightarrow$ 1.00 while predictable public-random and fixed hopping collapse to 0.0 (Fig. 3). This is the operational meaning of cryptographic unpredictability, and it is what a keyless pseudo-random hopper cannot provide.

The taxonomy also bounds the spread-spectrum baseline. DSSS is unaffected by every narrowband class, because none of them exceeds its processing-gain margin, but wideband barrage denies 70% of the band at once, drops DSSS throughput to 0.45, and takes its avoidance to 0.0. No hopping strategy fares better in that column: with 70% of the band denied, every hopper, cryptographic or otherwise, is reduced to 0.30. Barrage is therefore the one adversary in the study that defeats both families, which is the expected consequence of spreading jamming power across the entire band rather than concentrating it.

C. Learned versus cryptographic hopping at scale

The DQN, when it can be trained, beats cryptographic hopping on structured adversaries, since it learns to occupy clean

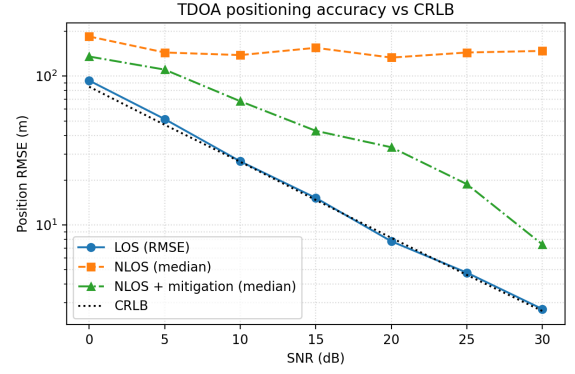


Fig. 2. Positioning error vs SNR. LOS RMSE hugs the CRLB; NLOS mitigation (median) approaches it as SNR grows.

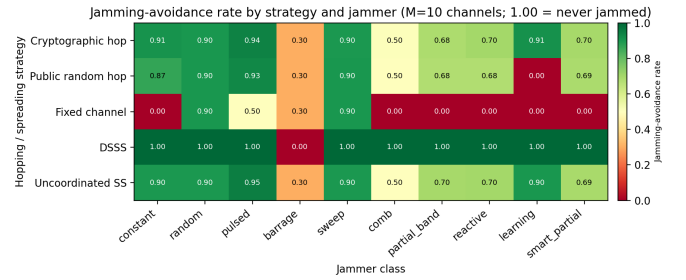


Fig. 3. Jamming-avoidance rate by strategy and jammer ($M=10$ channels; 1.00 = never jammed). DSSS is robust across every narrowband class and fails only against wideband barrage; fixed hopping fails broadly.

channels. Its advantage inverts as M scales. Under a fixed 20 s compute budget, avoidance falls from 0.91 at $M=10$ to 0.07 at $M=1000$, while training throughput collapses from 1,655 to 5 steps/s and the parameter count grows from 1.1×10^4 to 2.5×10^6 (Table II, Figs. 4–6). Cryptographic hopping is unaffected at 0.70 and needs no training.

Because a fixed wall-clock budget starves the wide-band models by construction, we repeat the experiment with the budget removed and the number of gradient updates held constant across all M . This second arm isolates the part of the degradation that belongs to the learning problem rather than to the allowance: equalising updates gives 0.90 at $M=10$ and 0.00 at $M=1000$. Fig. 6 reports the same effect as a cost, the training required to reach a fixed competence level of 0.90 avoidance, which is the quantity a deployment actually has to pay.

D. Securing the position, and its cost

Table III and Fig. 7 report attack-success rates. Cryptographic hopping alone reduces injection spoofing to $\approx 1/M$ but leaves replay at 1.0; adding a message signature stops spoofing yet a valid signature still loses to meaconing (replay 1.0), because it does not protect arrival-time integrity. The authenticated-timing layer drives both spoofing and replay to 0.0. Per decision, cryptographic hopping costs a single AES block (≈ 13 pJ) against 0.04–9.3 μ J for DL inference, a factor up to 7.2×10^5 , and it requires no training (Fig. 8).

TABLE II

DEEP LEARNING AT SCALE AGAINST AN ADAPTIVE 30% JAMMER, UNDER AN EQUAL-WALL-CLOCK BUDGET (10 SEEDS) AND EQUAL GRADIENT STEPS (3 SEEDS).

M	DQN (20 s)	DQN (equal steps)	Crypto	Train steps/s
10	0.91	0.90	0.70	1,655
100	0.84	0.83	0.70	181
1000	0.07	0.00	0.70	11
2320	0.16	0.00	0.70	5

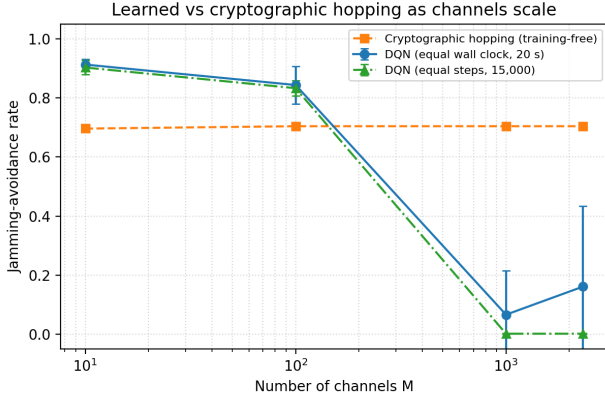


Fig. 4. Avoidance as channels scale, under both training regimes, against the training-free baseline.

Synchronisation tolerates a clock offset up to half the hop dwell (3.66 ms, 366 μ s, 36.6 μ s at 100/1000/10000 hops/s), with resync latencies of 80/8/0.8 ms (Fig. 9).

E. Statistical validation

All comparisons use independently seeded runs. Differences are assessed with the paired t -test and the non-parametric Wilcoxon signed-rank test [36], with 95% bootstrap confidence intervals and Cohen's d effect size [37], at $\alpha=0.05$ with Bonferroni correction over the reported family. Table IV confirms that cryptographic hopping significantly exceeds the learned model at $M=1000$ and the keyless baselines against the intelligent jammer, while the learned model significantly exceeds cryptographic hopping at $M=10$, quantifying the crossover rather than asserting a single winner.

VII. DISCUSSION

A. Boundary conditions

The findings are boundary-conditioned. A learned policy is the better anti-jammer *when* the channel count is small and training is affordable and *when* the adversary is not itself adaptive to the policy; it exploits structure that a blind keyed hopper ignores. Cryptographic hopping wins *when* the band is wide (its collision probability $\rightarrow 0$ as M grows), *when* training is impractical, and *when* the adversary is intelligent, because unpredictability is a property of key secrecy, not of data. Only the cryptographic path additionally secures the *position*: combined with authenticated timing it defeats jamming,

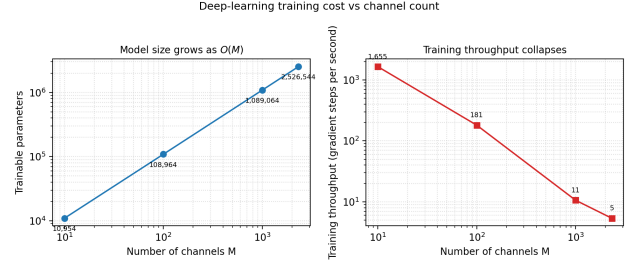


Fig. 5. $O(M)$ growth in parameters and the corresponding collapse in training throughput.

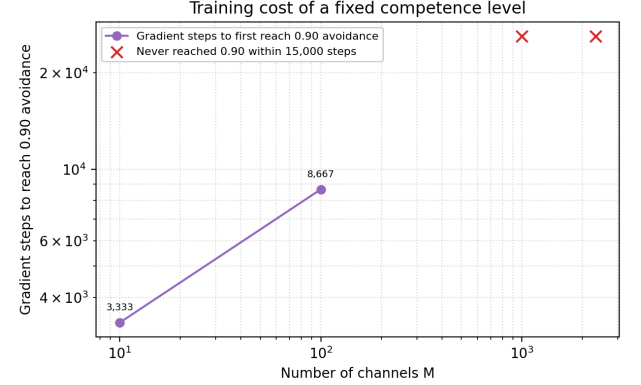


Fig. 6. Training cost of reaching a fixed competence level (0.90 avoidance) as the channel count grows.

spoofing, and replay together, which a communication-only learned hopper does not attempt.

B. Limits of the cryptographic approach

The cost analysis settles one common objection, that cryptography is as expensive as deep learning: generating a hop index is a single AES block, orders of magnitude below a neural forward pass and free of any training. The real costs are operational rather than computational, and they are not free. *Key distribution* is a logistical burden: keys must be provisioned before the mission and re-provisioned when compromised, and the pre-shared assumption of Section IV-B is exactly what an adversary with physical access attacks. *Key secrecy is a single point of failure*: a disclosed key converts the unpredictable hopper into the predictable one, which the learning jammer drives to 0.0 avoidance. *Synchronisation must be maintained*, and a drone that loses it pays the resynchronisation latency of Section VI before it can receive at all. Finally, the cryptographic hopper is *blind*: it cannot exploit exploitable structure in the jammer, which is precisely why the learned model wins at small M . Neither approach answers wideband barrage, which defeats both.

C. Threats to validity

Construct validity. The jammer-hopper interaction is simulated at the channel-occupancy level; a signal-level GCC-PHAT check validates the range-noise model, but full RF

TABLE III
ATTACK-SUCCESS RATE BY SECURITY LAYER ($M=100$). A VALID SIGNATURE DOES NOT STOP REPLAY; AUTHENTICATED TIMING DOES.

Layer	Spoofing	Replay
None	1.00	1.00
Crypto hopping only	0.01	1.00
Crypto + signature only	0.00	1.00
Crypto + authenticated timing	0.00	0.00

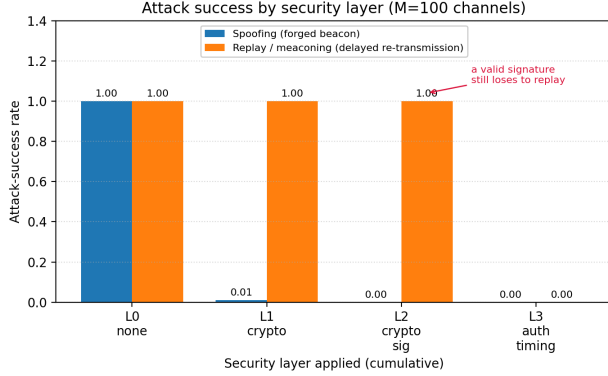


Fig. 7. Attack success by security layer.

propagation and hardware imperfections are abstracted. Two consequences are worth stating plainly. The DSSS baseline is modelled by a processing-gain tolerance and therefore represents an upper bound on what a real DSSS receiver achieves. Conversely, the barrage jammer pays no penalty in this model for spreading its power across the whole band, whereas a real barrage adversary trades per-channel power density for coverage; the barrage column is therefore pessimistic for the defenders and should be read as a bound, not a prediction. *Internal validity.* The scaling result is reported under both a fixed wall-clock budget and a fixed number of gradient updates precisely because the first alone would confound the budget with the learning problem; the $O(M)$ growth in parameters and per-step cost is architectural and holds under either. *External validity.* Results are simulation-based with a single beacon geometry and a ten-class taxonomy; field behaviour may differ, and hardware-in-the-loop validation remains future work. *Conclusion validity.* Multiple seeds, paired significance tests, confidence intervals, and Bonferroni correction guard against chance findings; NLOS metrics use medians because the error is heavy-tailed.

VIII. CONCLUSION

We benchmarked a reproduced deep-Q anti-jamming model against training-free cryptographic frequency hopping for GPS-denied drone TDOA positioning, under one environment, dataset, and metric suite. The learned model matches or beats cryptographic hopping at small channel counts but degrades as channels scale, under both an equal-wall-clock and an equal-gradient-step regime, whereas cryptographic hopping stays effective, training-free, and unpredictable, and, with an authenticated-timing layer, secures the position against

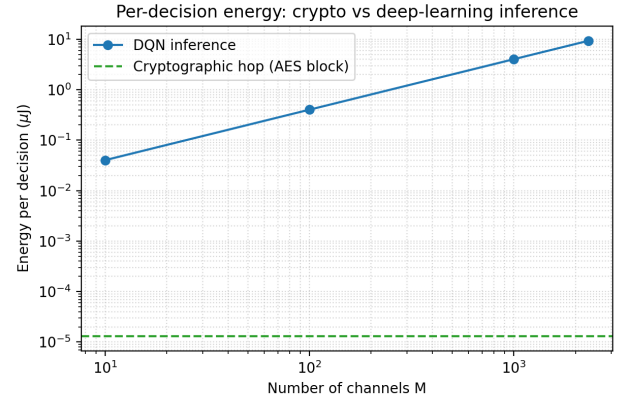


Fig. 8. Per-decision energy: cryptographic hop generation versus deep-learning inference.

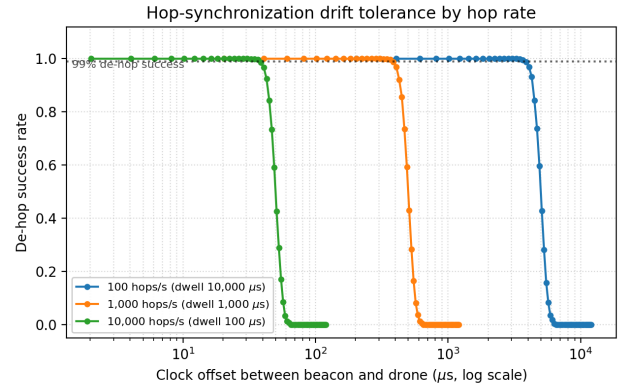


Fig. 9. Hop-synchronisation drift tolerance and resynchronisation latency versus hop rate.

jamming, spoofing, and replay together at pico-joule cost. The maximum-likelihood TDOA estimator reaches the CRLB, and NLOS mitigation recovers most of the loss. Extending the taxonomy to wideband barrage bounds both families: it is the one adversary that defeats spread spectrum and hopping alike. We release the datasets, code, and benchmark. Future work extends the study to signal-level propagation, hardware-in-the-loop trials, and adaptive-adversary co-training.

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TABLE IV
SIGNIFICANCE OF HEADLINE COMPARISONS (PAIRED t -TEST,
BONFERRONI CORRECTED). POSITIVE MEAN DIFFERENCE FAVOURS THE
FIRST METHOD.

Comparison	Condition	diff	p	d
crypto > DQN	$M=1000$	+0.61	< 0.001	3.29
DQN > crypto	$M=10$	+0.22	$< 10^{-9}$	24.3
crypto > random	learning jam.	+0.99	< 0.001	$\gg 1$

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